

JavaScript Interview Notes

Polyfill

A polyfill is a piece of code (usually JavaScript) that provides modern functionality on older browsers that do not support it.

```
if (!Array.prototype.includes) {
  Array.prototype.includes = function(search, start) {
    if (this == null) throw new TypeError('"this" is null or not defined');
    let arr = Object(this);
    let len = arr.length >>> 0;
    if (len === 0) return false;

    let i = start | 0;
    if (i < 0) i = Math.max(len + i, 0);

    while (i < len) {
      if (arr[i] === search || (Number.isNaN(arr[i]) && Number.isNaN(search))) {
        return true;
      }
      i++;
    }
    return false;
  };
}
```

First-Class Functions

JavaScript treats functions as first-class citizens. They can be assigned to variables, passed as arguments, returned from other functions, and stored in objects/arrays.

// Example: Passing function as argument

```
function processUserInput(callback) {
  const name = "Alisha";
  callback(name);
}
processUserInput(function(name) {
  console.log("Welcome " + name);
});
```

Function Hoisting Example

```
function check() {
  console.log("something..");
}
var check;
console.log(check);
// Output: function definition, because function declaration is hoisted before var declaration.
```

ES6 Features

// let and const

```

let x = 10;
const y = 20;

// Arrow functions
const add = (a, b) => a + b;

// Template literals
let name = "Alisha";
console.log(`Hello, ${name}!`);

// Destructuring
const [a, b] = [1, 2];
const {age} = {name: "Alisha", age: 22};

```

Slice vs Splice

```

let arr = [1,2,3,4,5];

console.log(arr.slice(1,3)); // [2,3]
console.log(arr);           // [1,2,3,4,5]

arr.splice(2,2,99);
console.log(arr); // [1,2,99,5]

```

Map vs ForEach

```

let arr = [1,2,3];

arr.forEach(x => console.log(x*2)); // just logs

let doubled = arr.map(x => x*2);
console.log(doubled); // [2,4,6]

```

Array Methods

```

let nums = [1,2,3,4];

// push, pop
nums.push(5); // [1,2,3,4,5]
nums.pop();   // [1,2,3,4]

// map, filter, reduce
let doubled = nums.map(n => n*2); // [2,4,6,8]
let evens = nums.filter(n => n%2===0); // [2,4]
let sum = nums.reduce((a,b) => a+b, 0); // 10

```

parseInt vs Number

```

parseInt("42px"); // 42
Number("42px");  // NaN

parseInt("3.14"); // 3
Number("3.14");   // 3.14

```

```
parseInt("abc123"); // NaN
Number("abc123");   // NaN
```

Map Callback Undefined Example

```
console.log([1,2,3,4,5].map(e => {
  if (e > 0) return;
  return e * 10;
}));
// Output: [undefined, undefined, undefined, undefined, undefined]
```

Session Storage vs Local Storage

```
// Local Storage
localStorage.setItem("name", "Alisha");
console.log(localStorage.getItem("name")); // "Alisha"

// Session Storage
sessionStorage.setItem("token", "12345");
console.log(sessionStorage.getItem("token")); // "12345"
```

WeakSet vs WeakMap

```
let ws = new WeakSet();
let obj1 = {name: "Alisha"};
ws.add(obj1);
console.log(ws.has(obj1)); // true

let wm = new WeakMap();
let user = {id:1};
wm.set(user, "data");
console.log(wm.get(user)); // "data"
```

Common Interview Questions

- What is a polyfill and when do you need it?
- Difference between map() and forEach().
- How does slice() differ from splice()?
- Explain localStorage vs sessionStorage.
- What are WeakSet and WeakMap? Why are they called 'weak'?
- parseInt vs Number vs parseFloat differences.
- Name 5 important ES6 features.